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Title of chapter: Chapter 0005 – Causality

Summary:

This chapter examines how causal reasoning extends beyond mere observation to answer why and what if questions about the world. It begins by demonstrating the limitations of observational data via scenarios such as the Berkeley graduate admissions case, within which aggregate statistics wrongfully displayed gender bias. The chapter presents structural causal models (SCMs) and causal graphs as formal tools for distinguishing between correlation and causation, describing the mechanisms by which variables impact one another, and assessing the consequences of interventions using the do-operator. Key graphical concepts such as forks, chains, and colliders show whether controlling for a variable clarifies or distorts causal inference. The backdoor criteria is a guideline for recognizing causal effects and deciding which variables need to be modified to eliminate confounding bias.

The chapter goes on to further explain counterfactual reasoning, mediation analysis, and its implications on fairness and discrimination. Counterfactuals allow us to measure direct and indirect consequences, providing a better understanding of whether a process or decision is biased. However, the chapter warns that causal reasoning in social contexts (e.g., race, gender, religion) need careful modeling since these categories are socially constructed and may not behave as stable, manipulable causes. Finally, causality provides a systematic approach to think about interventions and fairness, but it relies on well-defined models, valid assumptions, and a careful interpretation of what constitutes a fair or meaningful causal pathway.

What are two points that you learned after reading this chapter?:

- Observational correlations can be inaccurate because they do not represent how variables react to interventions; causal inference necessitates defining mechanisms through causal graphs or SCMs.
- Counterfactual reasoning enables us to quantify path-specific causal effects (such as direct and indirect discrimination), but its accuracy is dependent on the quality and validity of the underlying causal model.

What are two questions that you have about this chapter?:

- Given the socially constructed nature and context-dependence of complex social categories such as race and gender, how can we construct reliable causal models?
- In real-world fairness applications, how should we determine whether indirect impacts (e.g., education or socioeconomic status) are legitimate or discriminatory as pathways?

Do you have some initial thoughts or insights to approach the questions?:

- To model complex social variables, we may require multidisciplinary techniques that integrate sociological theory, domain expertise, and data-driven insights to construct meaningful causal structures, rather than depending on just statistical assumptions.

- In addition to causal analysis, determining whether indirect effects are discriminatory will most likely need ethical and legal judgment. Before evaluating whether causal routes are acceptable or unfair, we need to identify the objective of the fairness assessment: whether it is to detect bias, develop fair policies, or to ensure equitable outcomes.